Dr Vanessa Echeverria

Senior Lecturer, RMIT University, Melbourne, 3000, VIC
vanessa.echeverria@rmit.edu.au | https://www.vanessaecheverria.net/
ORCID: 0000-0002-2022-9588
Google Scholar: https://scholar.google.com/citations?user=X3YWKc8AAAAJ&hl=en

APPOINTMENTS

Senior Lecturer (Assistant Professor) | Nov 2026–present
School of Computing Technologies, RMIT University
VIC, Australia

Lecturer (Assistant Professor) | Jan 2025–Oct 2026
School of Computing Technologies, RMIT University
VIC, Australia

Postdoctoral Research Fellow | July 2022 – Jan 2025
Department of Human-Centred Computing, Faculty of Information Technology, Monash University
VIC, Australia

Assistant Professor | September 2020 – June 2022
Faculty of Electrical and Computing Engineering (FIEC), Escuela Superior Politécnica del Litoral (ESPOL)
Guayaquil, Ecuador

Postdoctoral Research Fellow | January 2020 – August 2020
Human-Computer Interaction Institute, Carnegie Mellon University
Pittsburgh, PA, United States

Assistant Professor | October 2013 – December 2016
Faculty of Electrical and Computing Engineering (FIEC), Escuela Superior Politécnica del Litoral (ESPOL)
Guayaquil, Ecuador

EDUCATION

PhD in Learning Analytics | 2016–2020
Connected Intelligence Centre
University of Technology Sydney, Sydney, Australia

M Sc. in Artificial Intelligence | 2013
Major in Engineering and Computer Science
Cum Laude
Department of Computer Science
KU Leuven, Leuven, Belgium

B.S. in Computer Science | 2004–2010
Major in Multimedia Systems
Faculty of Electrical and Computing Engineering (FIEC)
Escuela Superior Politécnica del Litoral, Guayaquil, Ecuador (ESPOL), Guayaquil, Ecuador

HONORS / AWARDS

Early Career Researcher Award, Society of Learning Analytics Research | 2025
Dean's Research Team of the Year Award, Monash University, Centre for Learning Analytics at Monash (CoLAM) | 2024
Runner-up for the 2023 Best Paper Award by IEEE Pervasive Computing, IEEE Computer Society Publications Board | 2024
Runner-up Best Paper Award, Learning Analytics and Knowledge International Conference | 2024
Best Design Paper Award, International Conference on Computer Supported Collaborative Learning | 2022
Runner-up Best Paper Award, European Conference of Technology Enhanced Learning | 2020
Chancellor's List of Outstanding PhD. Theses, Honourable Mention. University of Technology Sydney (UTS), Australia | 2020
Best Paper Award, International Conference on Artificial Intelligence in Education | 2020

Travel Grant Award, APSCE Merit Scholarship, International Conference on Computers in Education, Christchurch, NZ ($500) | 2017
Best Paper Award, IEEE Ecuador Technical Chapters Meeting | 2016

GRANTS

Transforming clinical performance feedback with expert-driven AI-powered analytics
Boundary Crosser Scheme - STEM College – RMIT University
2026 | 49,160 AUD

Career Reignite program – STEM College – RMIT University
2025 | 19,896 AUD

Augmenting the Debrief of Emergency Resuscitation In-situ Simulation utilising Artificial Intelligence
Laerdal Foundation
R. Martinez-Maldonado (PI**), V. Echeverria (Co-PI)**, Connell (Co-PI), J. Davis (Co-PI), S. Sevenhuysen (Co-PI)
2024 | 72,058 AUD

Visually communicating catchment water quality information for diverse stakeholders: A case study of Macalister Irrigation District. S**.**
ECA Seed Grant Monash University
Goodwin (PI), V. Echeverria (Co-PI), Anna Lintern (Co-PI)
2024 | 22,948 AUD

INVITED GUEST LECTURES

Data Visualizations and Narratives | Fall 2018 - Fall 2017
Faculty of Transdisciplinary Innovation, University of Technology Sydney (UTS), Sydney, AUS

Human-Centered Data Visualizations | Fall 2019
Faculty of Transdisciplinary Innovation, University of Technology Sydney (UTS), Sydney, AUS

PUBLICATIONS

All publications can be found on my Google Scholar page:
https://scholar.google.com/citations?user=X3YWKc8AAAAJ&hl=en

JOURNALS

J26 Milesi, M. E., Alfredo, R., **Echeverria, V.,** Yan, L., Zhao, L., Jin, Y., ... & Martinez-Maldonado, R. (2026). From comic panels to clinical practice: data comics as a learning analytics tool in nursing simulation. Smart Learning Environments, 13(1), 14.

J25 Fernandez-Nieto, G. M., **Echeverria, V.**, Li, Y., Tsai, Y. S., Sha, L., Chen, G., ... & Swiecki, Z. (2026). Capturing and sharing know-how through visual process representations: a human-Centred approach to teacher workflows. *Behaviour & Information Technology*, 1-30.

J24 Zhao, L., **Echeverria, V.,** Tan, Y., Yan, L., Li, X., Alfredo, R., ... & Martinez-Maldonado, R. (2026). Ordered Networked Analysis of Multimodal Data in Healthcare Simulations: Dissecting Team Communication Tactics. *International Journal of Artificial Intelligence in Education*, 100011.

J23 Alfredo, R., Milesi, M., **Echeverria, V**., Gašević, D., Buckingham Shum, S., Zhao, L., ... & Martinez-Maldonado, R. (2025). Co-designing AI-powered learning analytics: bringing students and teachers together. *International Journal of Educational Technology in Higher Education*, *22*(1), 78.

J22 Yan, L., Martinez-Maldonado, R., Jin, Y., **Echeverria, V.,** Milesi, M., Fan, J., ... & Gašević, D. (2025). The effects of generative AI agents and scaffolding on enhancing students' comprehension of visual learning analytics. *Computers & Education*, 105322.

J21 Jin, Y., Yan, L., **Echeverria, V**., Gašević, D., & Martinez-Maldonado, R. (2025). Generative AI in higher education: A global perspective of institutional adoption policies and guidelines. *Computers and Education: Artificial Intelligence*, *8*, 100348.

J20 Yan, L., Gašević, D., **Echeverria, V.**, Zhao, L., Jin, Y., Li, X., & Martinez-Maldonado, R. (2025). In Sync or Out of Sync? Understanding Stress and Learning Performance in Collaborative Healthcare Simulations through Physiological Synchrony and Arousal. *International Journal of Artificial Intelligence in Education*, 1-32. https://doi.org/10.1007/s40593-025-00475-9

J19 Yan, L., Martinez-Maldonado, R., Jin, Y., **Echeverria, V**., Milesi, M., Fan, J., ... & Gašević, D. (2025). The effects of generative AI agents and scaffolding on enhancing students' comprehension of visual learning analytics. *Computers & Education*, 105322. https://doi.org/10.1016/j.compedu.2025.105322

J18 Pozdniakov, S., Martinez-Maldonado, R., Tsai, Y.-S., **Echeverria, V.,** Swiecki, Z., & Gasevic, D. (2025). Investigating the Effect of Visualization Literacy and Guidance on Teachers' Dashboard Interpretation. *Journal of Learning Analytics*, 1-25. https://doi.org/10.18608/jla.2024.8471

J17 Jin, Y., Yan, L., **Echeverria, V.,** Gašević, D., & Martinez-Maldonado, R. (2025). Generative AI in higher education: A global perspective of institutional adoption policies and guidelines. Computers and Education: Artificial Intelligence, 8, 100348. https://doi.org/10.1016/j.caeai.2024.100348

J16 **Echeverria, V.,** Nieto, G. F., Zhao, L., Palominos, E., Srivastava, N., Gašević, D., ... & Martinez-Maldonado, R. (2025). A learning analytics dashboard to support students' reflection on collaboration. *Journal of Computer Assisted Learning*, *41*(1), e13088. https://doi.org/10.1111/jcal.13088

J15 Jin, Y., **Echeverria, V.,** Yan, L., Zhao, L., Alfredo, R., Tsai, Y.-S., Gašević, D., & Martinez-Maldonado, R. (2024). FATE in MMLA: A Student-Centred Exploration of Fairness, Accountability, Transparency, and Ethics in Multimodal Learning Analytics. *Journal of Learning Analytics*, *11*(3), 6-23. https://doi.org/10.18608/jla.2024.8351

J14 Alfredo, R., **Echeverria, V.,** Zhao, L., Lawrence, L., Fan, J. X., Yan, L., ... & Martinez-Maldonado, R. (2024). Designing a Human-Centred Learning Analytics Dashboard In-Use. *Journal of Learning Analytics*, *11*(3), 62-81. https://doi.org/10.18608/jla.2024.8487

J13 Yan, L., **Echeverria, V.,** Jin, Y., Fernandez-Nieto, G., Zhao, L., Li, X., ... & Martinez-Maldonado, R. (2024). Evidence-based multimodal learning analytics for feedback and reflection in collaborative learning. *British Journal of Educational Technology*, *55*(5), 1900-1925. https://doi.org/10.1111/bjet.13498

J12 Alfredo, R., **Echeverria, V.,** Jin, Y., Yan, L., Swiecki, Z., Gašević, D., & Martinez-Maldonado, R. (2024). Human-centred learning analytics and AI in education: A systematic literature review. *Computers and Education: Artificial Intelligence*, 100215. https://doi.org/10.1016/j.caeai.2024.100215

J11 Lawrence, L., **Echeverria, V.,** Yang, K., Aleven, V., & Rummel, N. (2024). How teachers conceptualise shared control with an AI co-orchestration tool: A multiyear teacher-centred design process. *British Journal of Educational Technology*, *55*(3), 823-844. https://doi.org/10.1111/bjet.13372

J10 Martinez-Maldonado, R., **Echeverria, V.,** Fernandez-Nieto, G., Yan, L., Zhao, L., Alfredo, R., ... & Gašević, D. (2023) Lessons Learnt from a Multimodal Learning Analytics Deployment In-the-wild. *ACM Transactions on Computer-Human Interaction*. https://doi.org/10.1145/3622784

J9 **Echeverria, V.,** Yang, K., Lawrence, L., Rummel, N., Aleven, V. (2023). Designing Hybrid Human–AI Orchestration Tools for Individual and Collaborative Activities: A Technology Probe Study. *IEEE Transactions on Learning Technologies*, vol. 16, no. 2, pp. 191-205, https://doi.org.10.1109/TLT.2023.3248155

J8 **Echeverria, V.,** Martinez-Maldonado, R., Yan, L., Zhao, L., Fernandez-Nieto, G., Gašević, D., & Shum, S. B. (2023). HuCETA: A Framework for Human-Centered Embodied Teamwork Analytics. *IEEE Pervasive Computing.* vol. 22, no. 1, pp. 39-49, doi: 10.1109/MPRV.2022.3217454

J7 Fernandez-Nieto, G. M., **Echeverria, V.,** Buckingham Shum, S., Mangaroska, K., Kitto, K., Palominos, E., Axisa, C., Martinez-Maldonado, R. (2021). Storytelling With Learner Data: Guiding Student Reflection on Multimodal Team Data. *IEEE Transactions on Learning Technologies*. https://doi.org/10.1109/TLT.2021.3131842

J6 Martinez-Maldonado, R., Elliott, D., Axisa, C., Power, T., **Echeverria, V.,** & Buckingham Shum, S. (2022). Designing translucent learning analytics with teachers: An elicitation process. *Interactive Learning Environments*, 30(6), 1077-1091. https://doi.org/10.1080/10494820.2019.1710541

J5 Fernandez-Nieto, G., Martinez-Maldonado R., **Echeverria, V.,** Kitto, K., An, P., & Buckingham-Shum, S. (2021). What Can Analytics for Teamwork Proxemics Reveal About Positioning Dynamics in Clinical Simulations? *Proceedings of the ACM on Human-Computer Interaction – CSCW 2021*, 2: 1-24. https://doi.org/10.1145/3449284

J4 ☆ Martinez-Maldonado, R., **Echeverria, V.**, Mangaroska, K. et al. (2021). Moodoo the Tracker: Spatial Classroom Analytics for Characterising Teachers' Pedagogical Approaches. *International Journal of Artificial Intelligence in Education*. https://doi.org/10.1007/s40593-021-00276-w

J3 ☆ Martinez-Maldonado, R., Gašević, D., **Echeverria, V.,** Nieto, G. F., Swiecki, Z., & Shum, S. B. (2021). What do you mean by collaboration analytics? A conceptual model. *Journal of Learning Analytics*, 8(1), 126-153.

J2 Martinez-Maldonado, R., Schulte, J., **Echeverria, V**., Gopalan, Y. and Buckingham Shum, S. (2020). Where is the Teacher? Digital Analytics for Classroom Proxemics. *Journal of Computer Assisted Learning, JCAL*, 36(5): 741-762.

J1 **Echeverria, V.,** Martinez-Maldonado, R., Buckingham Shum, S., Granda, R., Chiluiza, K., Conati, C. (2018). Exploratory versus Explanatory Visual Learning Analytics: Driving Teachers' Attention through Educational Data Storytelling. *Journal of Learning Analytics*, JLA, 5(2): 72-97.

CONFERENCES [STRICTLY PEER-REVIEWED]

C43 Falhs, A. C., Borchers, C., **Echeverria, V.,** Yang, K., Rummel, N., & Aleven, V. (2025, September). How Expertise Levels Shape Preferences and Reflection Needs: Towards AI Reflection Systems for Teacher Empowerment. In *European Conference on Technology Enhanced Learning* (pp. 120-125). Cham: Springer

C42 Milesi, M. E., Mejia-Domenzain, P., Brandl, L., **Echeverria, V.,** Jin, Y., Gasevic, D., ... & Martinez-Maldonado, R. (2025, April). " Piecing Data Connections Together Like a Puzzle": Effects of Increasing Task Complexity on the Effectiveness of Data Storytelling Enhanced Visualisations. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (pp. 1-18). **Ranking: Core A***

C41 **Echeverria, V.,** Zhao, L., Alfredo, R., Milesi, M. E., Jin, Y., Abel, S., ... & Martinez-Maldonado, R. (2025, April). TeamVision: An AI-powered Learning Analytics System for Supporting Reflection in Team-based Healthcare Simulation. In *Proc. of the 2025 CHI Conf. on Human Factors in Computing Systems* (pp. 1-22). **Ranking: Core A***

C40 Prieto, L. P., Alfredo, R., Díaz-Chavarría, H. B., Martinez-Maldonado, R., & Echeverría, V. (2025, March). VALA/AID: A method for rapid, participatory value-sensitive learning analytics and artificial intelligence design. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference* (pp. 950-956). **Ranking: Core A**

C39 Alfredo, R., Mejia-Domenzain, P., **Echeverria, V.,** Rahayu, D., Zhao, L., Alajlan, H., ... & Martinez-Maldonado, R. (2025, March). TeamTeachingViz: Benefits, Challenges, and Ethical Considerations of Using a Multimodal Analytics Dashboard to Support Team Teaching Reflection. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference* (pp. 58-69). **Ranking: Core A**

C38 Yan, L., Gasevic, D., **Echeverria, V.**, Jin, Y., Zhao, L., & Martinez-Maldonado, R. (2025). From Complexity to Parsimony: Integrating Latent Class Analysis to Uncover Multimodal Learning Patterns in Collaborative Learning. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference* (pp. 70-81). https://doi.org/10.1145/3706468.3706476 **Ranking: Core A**

C37 Jin, Y., Yang, K., Yan, L., **Echeverria, V.,** Zhao, L., Alfredo, R., ... & Martinez-Maldonado, R. (2025). Chatting with a learning analytics dashboard: The role of generative AI literacy on learner interaction with conventional and scaffolding chatbots. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference* (pp. 579-590). https://doi.org/10.1145/3706468.3706545 **Ranking: Core A**

C36 Yang, K. B., Borchers, C., Falhs, A. C., **Echeverria, V.,** Karumbaiah, S., Rummel, N., & Aleven, V. (2024, September). Leveraging Multimodal Classroom Data for Teacher Reflection: Teachers' Preferences, Practices, and Privacy Considerations. In *European Conference on Technology Enhanced Learning* (pp. 498-511). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-72315-5_34

C35 Yan, L., Zhao, L., **Echeverria, V.,** Jin, Y., Alfredo, R., Li, X., ... & Martinez-Maldonado, R. (2024, July). VizChat: enhancing learning analytics dashboards with contextualised explanations using multimodal generative

AI chatbots. In *International Conference on Artificial Intelligence in Education* (pp. 180-193). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-64299-9_13 **Ranking: Core A | acceptance rate: 23%**

C34 Yan, L., **Echeverria, V.,** Fernandez-Nieto, G. M., Jin, Y., Swiecki, Z., Zhao, L., ... & Martinez-Maldonado, R. (2024, May). Human-AI Collaboration in Thematic Analysis using ChatGPT: A User Study and Design Recommendations. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1-7). https://doi.org/10.1145/3613905.3650732

C33 Milesi, M. E., Alfredo, R., **Echeverria, V.,** Yan, L., Zhao, L., Tsai, Y. S., & Martinez-Maldonado, R. (2024, May). " It's Really Enjoyable to See Me Solve the Problem like a Hero": GenAI-enhanced Data Comics as a Learning Analytics Tool. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1-7). https://doi.org/10.1145/3613905.3651111

C32 Shao, H., Martinez-Maldonado, R., **Echeverria, V.,** Yan, L., & Gasevic, D. (2024, May). Data Storytelling in Data Visualisation: Does it Enhance the Efficiency and Effectiveness of Information Retrieval and Insights Comprehension?. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (pp. 1-21). https://doi.org/10.1145/3613904.3643022 **Ranking: Core A* | acceptance rate: 26.3%**

C31 Zhao, L., **Echeverria, V.,** Swiecki, Z., Yan, L., Alfredo, R., Li, X., ... & Martinez-Maldonado, R. (2024, March). Epistemic Network Analysis for End-users: Closing the Loop in the Context of Multimodal Analytics for Collaborative Team Learning. In *Proc of the 14th Learning Analytics and Knowledge Conference* (pp. 90-100). https://doi.org/10.1145/3636555.3636855 **Ranking: Core A | acceptance rate: 32.7%**

C30 Fernandez-Nieto, G. M., Martinez-Maldonado, R., **Echeverria, V.,** Kitto, K., Gašević, D., & Buckingham Shum, S. (2024, March). Data storytelling editor: A teacher-centred tool for customising learning analytics dashboard narratives. In *Proceedings of the 14th Learning Analytics and Knowledge Conference* (pp. 678-689). https://doi.org/10.1145/3636555.3636930 **Ranking: Core A | acceptance rate: 32.7%**

C29 **Echeverria, V.**, Yan, L., Zhao, L., Abel, S., Alfredo, R., Dix, S., ... & Martinez-Maldonado, R. (2024). TeamSlides: a Multimodal Teamwork Analytics Dashboard for Teacher-guided Reflection in a Physical Learning Space. *In LAK24: 14th International Learning Analytics and Knowledge Conference*, ACM. https://doi.org/10.1145/3636555.3636857 **Ranking: Core A | acceptance rate: 32.7%**
Runner-up best paper award

C28 *Alfredo, R., **Echeverria, V.,** Jin, Y., Swiecki, Z., Gašević, D., & Martinez-Maldonado, R. (2024). SLADE: A Method for Designing Human-Centred Learning Analytics Systems. *In LAK24: 14th Int. Learning Analytics and Knowledge Conf*, ACM. Vol. 10, No. 3636555.3636847. **Ranking: Core A | acceptance rate: 32.7%**

C27 Lawrence, L., **Echeverria, V.,** & Rummel, N. (2023). Acknowledging Power Structures in the Design of AI-Based Orchestration Tools. *In Proceedings of the 16th International Conference on Computer-Supported Collaborative Learning-CSCL 2023*, pp. 173-176. International Society of the Learning Sciences.

C26 Chiluiza, K., **Echeverria, V.,** Pinargote, A*., Carrillo, G., Ortiz-Rojas, M. (2023). Implementing Synchronous Hybrid Learning: Insights from Teachers and Students' Experiences. In: Berrezueta, S. (eds) *Proceedings of the 18th Latin American Conference on Learning Technologies (LACLO 2023). LACLO 2023*. Lecture Notes in Educational Technology. Springer, Singapore. https://doi.org/10.1007/978-981-99-7353-8_6

C25 Ochoa, X., **Echeverria, V.,** Carrillo, G., Heredia, V., & Chiluiza, K. (2023). Supporting Online Collaborative Work at Scale: A Mixed-Methods Study of a Learning Analytics Tool. *In Proceedings of the Tenth ACM Conference on Learning@ Scale* , ACM, New York, NY, USA, 237–247. https://doi.org/10.1145/3573051.3596165 **acceptance rate: 28%**

C24 Núnez, A. G., **Echeverria, V.,** Zuniga-Prieto, M., Auria, B., & De Laet, T. (2023). TMBQ-LT: A Student-Facing Learning Tool to Support Time Management Skills. *In Proceedings of the 15th International Conference on Computer Supported Education (CSEDU 2023)* - Volume 1, pages 217-224. DOI: https://10.5220/0011847500003470

C23 Yang, K. B, **Echeverria, V.**, Lu, Z., Mao, H., Holstein, K., Rummel, N., Aleven, V. (2023). Pair-Up: Prototyping Human-AI Co-orchestration of Dynamic Transitions between Individual and Collaborative Learning in the Classroom. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23).* Association for Computing Machinery, New York, NY, USA, Article 453, 1–17. https://doi.org/10.1145/3544548.3581398 **Ranking: Core A* | acceptance rate: 27.6%**

C22 Pozdniakov, S., Martinez-Maldonado, R., Tsai, Y. S., **Echeverria, V.,** Srivastava, N., & Gasevic, D. (2023). How Do Teachers Use Dashboards Enhanced with Data Storytelling Elements According to their Data Visualisation Literacy Skills?. *LAK23: 13th International Learning Analytics and Knowledge Conference*. Association for Computing Machinery, New York, NY, USA, 89–99. https://doi.org/10.1145/3576050.3576063 **Ranking: Core A | acceptance rate: 35.8%**

C21 Lawrence, L., Guo, B., **Echeverria, V.,** Kang, Z., Bathala, V., Li, C., ... & Rummel, N. (2022). Co-designing AI-based orchestration tools to support dynamic transitions: Design narratives through conjecture mapping. *In Proceedings of the 15th International Conference on Computer-Supported Collaborative Learning-CSCL 2022.* International Society of the Learning Sciences. **Best paper award**

C20 Yang, K. B., Lu, Z., **Echeverria, V.,** Sewall, J., Lawrence, L., Rummel, N., & Aleven, V. (2022). Technology Ecosystem for Orchestrating Dynamic Transitions Between Individual and Collaborative AI-Tutored Problem Solving. *Artificial Intelligence in Education: 23rd International Conference, AIED 2022, Lecture Notes in Computer Science*, vol 13355. Springer, Cham. https://doi.org/10.1007/978-3-031-11644-5_66 **Ranking: Core A | acceptance rate: 20%**

C19 Silva, I., Wong, A., Auria, B., Zambrano, D., & **Echeverria, V.** (2022). Gamification in Engineering Education: Exploring Students' Performance, Motivation, and Engagement. *2022 IEEE Sixth Ecuador Technical Chapters Meeting (ETCM)*, Quito, Ecuador, 2022, pp. 1-6, doi: https://10.1109/ETCM56276.2022.9935729 **Best paper award**

C18 Avila, A. G. N., Feraud, I. F. S., Solano-Quinde, L. D., Zuniga-Prieto, M., **Echeverria, V.,** & De Laet, T. (2022). Learning Analytics to Support the Provision of Feedback in Higher Education: A Systematic Literature Review. In *2022 XVII Latin American Conference on Learning Technologies (LACLO)* (pp. 01-08). IEEE.

C17 **Echeverria, V.**, Villacres-Wong, M., Ochoa, X., Chiluiza, K. (2022). An Exploratory Evaluation of a Feedback Report, *LAK22: 12th International Learning Analytics and Knowledge Conference (LAK22).* Association for Computing Machinery, New York, NY, USA, 478–484. https://doi.org/10.1145/3506860.3506890 **Ranking: Core A | acceptance rate: 25.3%**

C16 Yang, K. B., Lawrence, L., **Echeverria, V.,** Guo, B., Rummel, N., & Aleven, V. (2021). Surveying Teachers' Preferences and Boundaries Regarding Human-AI Control in Dynamic Pairing of Students for Collaborative Learning. *Technology-Enhanced Learning for a Free, Safe, and Sustainable World. EC-TEL 2021. Lecture Notes in Computer Science*, vol 12884. Springer, Cham. https://doi.org/10.1007/978-3-030-86436-1_20.

C15 Jang, J., Lee, J., **Echeverria, V.,** Lawrence, L., & Aleven, V. (2021, April). Explorations of Designing Spatial Classroom Analytics with Virtual Prototyping. *In LAK21: 11th International Learning Analytics and Knowledge Conference (LAK21).* Association for Computing Machinery, New York, NY, USA, 518–524. https://doi.org/10.1145/3448139.3448192 **Ranking: Core A | acceptance rate: 29.3%**

C14 Barzola, V., Pichardo, H., Macías, J., Zambrano, D., & **Echeverria, V**. (2021). "I need more motivation": Engaging students in the gamification design process. In *Technology-Enhanced Learning for a Free, Safe, and Sustainable World: 16th European Conference on Technology Enhanced Learning, EC-TEL 2021, Bolzano, Italy, September 20-24, 2021, Proceedings 16* (pp. 310-314). Springer International Publishing.

C13 Yang, K. B., **Echeverria, V**., Wang, X., Lawrence, L., Holstein, K., Rummel, N., & Aleven, V. (2021). Exploring Policies for Dynamically Teaming up Students through Log Data Simulation. *14th International Conference on Educational Data Mining (EDM)* (Online)

C12 Auria, B., *Barzola, V., & **Echeverria, V**. (2021). A Collaborative Design Tool to Support Learning Environments During Emergent Remote Learning. In 2021 IEEE URUCON, Montevideo, Uruguay, 2021, pp. 541-546, doi: https://10.1109/URUCON53396.2021.9647346

C11 **Echeverria, V.**, Holstein, K., Huang, J., Sewall, J., Rummel, N., & Aleven, V. (2020). Exploring human–AI control over dynamic transitions between individual and collaborative learning. *In Addressing Global Challenges and Quality Education: 15th European Conference on Technology Enhanced Learning, EC-TEL 2020*, Heidelberg, Germany, September 14–18, 2020, Proceedings 15 (pp. 230-243). Springer, Cham. https://doi.org/10.1007/978-3-030-57717-9_17. **Runner-up best paper award**

C10 Martinez-Maldonado, R., **Echeverria, V.,** Fernandez Nieto, G. and Buckingham Shum, S. (2020). From Data to Insights: A Layered Storytelling Approach for Multimodal Learning Analytics. *Proceedings of the ACM CHI*

Conference on Human Factors in Computing Systems, (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–15. https://doi.org/10.1145/3313831.3376148
Ranking: Core A* | acceptance rate: 24.3% | FWCI: 6.29 | TOP 2% citations

C9 Martinez-Maldonado, R., **Echeverria, V**., Schulte, J., Shibani, A., Mangaroska, K. and Buckingham Shum, S. (2020). Moodoo: Indoor Positioning Analytics for Characterising Classroom Teaching. *Proceedings of the 21st International Conference on Artificial Intelligence in Education, AIED 2020*. Lecture Notes in Computer Science, vol 12163. Springer, Cham. https://doi.org/10.1007/978-3-030-52237-7_29
Ranking: Core A | acceptance rate: 26.6% **Best paper award**

C8 **Echeverria, V.,** Martinez-Maldonado, R. and Buckingham Shum, S. (2019). Towards Collaboration Translucence: Giving Meaning to Multimodal Group Data. *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, (CHI '19).* Association for Computing Machinery, New York, NY, USA, Paper 39, 1–16. https://doi.org/10.1145/3290605.3300269
Ranking: Core A* | acceptance rate: 23.8% | FWCI: 7.12 | TOP 2% citations

C7 Buckingham Shum, S., **Echeverria, V.** and Martinez-Maldonado, R. (2019). Making Sense of Collocated Team Activity: The Multimodal Matrix as a Quantified Ethnography Methodology. *Proceedings of 1st International Conference on Quantitative Ethnography, ICQE 2019*.

C6 Martinez-Maldonado, R., **Echeverria, V.,** Santos, A., Santos, O., and Yacef, K. (2018). Physical Learning Analytics: A Multimodal Perspective. In *Proceedings of the 8th International Conference on Learning Analytics and Knowledge (LAK '18).* Association for Computing Machinery, New York, NY, USA, 375–379. https://doi.org/10.1145/3170358.3170379 **Ranking: Core A | acceptance rate: 30%**

C5 **Echeverria, V.,** Martinez-Maldonado, R. Granda, R., Chiluiza, K., Conati, C., and Buckingham Shum, S. (2018). Driving Data Storytelling from Learning Design. In *Proceedings of the 8th International Conference on Learning Analytics and Knowledge (LAK '18).* Association for Computing Machinery, New York, NY, USA, 131–140. https://doi.org/10.1145/3170358.3170380 **Ranking: Core A | acceptance rate: 30%**

C4 **Echeverria, V.,** Martinez-Maldonado, R. Power, T., Hayes, C., and Buckingham Shum, S. (2018). Where is the Nurse? Towards Automatically Visualising Meaningful Team Movement in Healthcare Education. *Proceedings of the 19th International Conference on Artificial Intelligence in Education, AIED 2018.* Lecture Notes in Computer Science, vol 10948. Springer, Cham. https://doi.org/10.1007/978-3-319-93846-2_14
Ranking: Core A | acceptance rate: 23%

C3 **Echeverria, V.,** Martinez-Maldonado, R., Chiluiza, K., and Buckingham Shum, S. (2017). DBCollab: Automated feedback for face-to-Face Group Database Design. *Proceedings of the 25th International Conference on Computers in Education*, ICCE 2017.

C2 Falcones, G., Wong-Villacres, M., **Echeverria, V.,** & Garcia, K. C. (2016). Enhancing quality of argumentation in a co-located collaborative environment through a tabletop system. *2016 IEEE Ecuador Technical Chapters Meeting (ETCM),* Guayaquil, Ecuador, 2016, pp. 1-6, doi: 10.1109/ETCM.2016.7750832. **Best paper award**

C1 Domínguez, F., Chiluiza, K., **Echeverria, V.,** & Ochoa, X. (2015). Multimodal selfies: Designing a multimodal recording device for students in traditional classrooms. *In Proceedings of the 2015 ACM on International Conference on Multimodal Interaction (ICMI '15).* Association for Computing Machinery, New York, NY, USA, 567–574. https://doi.org/10.1145/2818346.2830606

WORKSHOPS, SYMPOSIUMS AND POSTERS [PEER-REVIEWED]

W9 Damşa, C., **Echeverria, V**., Popov, V., Chernikova, O., Karlgren, K., Milesi, M., ... & Eide, H. (2025). Enhancing Team-Based Medical Simulations: Learning Through Reflection with Analytics and AI Tools. In *Proceedings of the 18th International Conference on Computer-Supported Collaborative Learning-CSCL 2025, pp. 516-524*. International Society of the Learning Sciences.

W8 Fernández-Nieto, G. M., Echeverria, V., Martinez-Maldonado, R., & Shum, S. B. (2024). YarnSense: Automated data storytelling for multimodal learning analytics. In *Data Storytelling and Learning Analytics Workshop 2024* (pp. 124-138). CEUR-WS. https://ceur-ws.org/Vol-3667/DS-LAK24_paper_3.pdf

W7 Pinargote, A., Calderón, E., Cevallos, K., Carrillo, G., Chiluiza, K., & Echeverría, V. (2024). Automating Data Narratives in Learning Analytics Dashboards using GenAI. In *LAK Workshops* (pp. 150-161). https://ceur-ws.org/Vol-3667/DS-LAK24_paper_5.pdf

W6 Barzola, V., Alvarado, E., *Loja, C., *Velez, A., Silva, I., & **Echeverria, V**. (2022). Analyzing Speech Data to Detect Work Environment in Group Activities. In *International Conference on Artificial Intelligence in Education* (pp. 357-361). Cham: Springer International Publishing.

W5 Martinez-Maldonado, R., **Echeverria, V.**, Santos, A., Yacef, K., and Pechenizkiy, M. (2017) How to Capitalise on Mobility, Proximity and Motion Analytics to Support Formal and Informal Education? CEUR Workshop *Proceedings of the 2nd International Workshop on Learning Analytics across Digital and Physical Spaces (Cross-LAK)* – held at International Conference on Learning Analytics and Knowledge, LAK 2017.

W4 **Echeverria, V.,** Falcones, G., Castells, J., Granda, R., & Chiluiza, K. (2017). Multimodal collaborative workgroup dataset and challenges. CEUR *Workshop Proceedings of the 2nd International Workshop on Learning Analytics across Digital and Physical Spaces (Cross-LAK)* – held at International Conference on Learning Analytics and Knowledge, LAK 2017.

W3 **Echeverria, V.,** Domínguez, F., & Chiluiza, K. (2016). Towards a distributed framework to analyze multimodal data. CEUR Workshop Proceedings of the 1st International Workshop on Learning Analytics across Digital and Physical Spaces (Cross-LAK) – held at the 6th International Conference on Learning Analytics and Knowledge, LAK 2016.

W2 Wong-Villacres, M., Ortiz, M., **Echeverria, V.,** & Chiluiza, K. (2015). A tabletop system to promote argumentation in computer science students. *Proceedings of the 2015 International Conference on Interactive Tabletops & Surfaces*, ITS 2015.

W1 **Echeverria, V.,** Avendaño, A., Chiluiza, K., Vásquez, A., & Ochoa, X. (2014). Presentation skills estimation based on video and kinect data analysis. *Proceedings of the 2014 ACM workshop on Multimodal Learning Analytics Workshop and Grand Challenge* – held at the 16th ACM International Conference on Multimodal Interaction, ICMI 2014.

TUTORIALS AND WORKSHOPS ORGANISED

T5 Di Mitri, D., Srivastava, N., Fernandez Nieto, G., Echeverria, V., Cobos, R., Tudur Sadashiva, A., ... & Cukurova, M. (2025, July). The Second International Workshop on Multimodal Artificial Intelligence in Education (MAIEd'25). In *International Conference on Artificial Intelligence in Education* (pp. 292-299). Cham: Springer Nature Switzerland.

T4 Fernandez-Nieto, G. M., **Echeverria, V.,** Martinez-Maldonado, R., Tsai, Y., Lawrence, L., Singh, S., Pozdniakov, S., Chen, L. K., Gong, J., & Yarnall, L. (2024, March). *Data storytelling narratives and learning analytics*. In *Companion Proceedings of the 14th International Conference on Learning Analytics & Knowledge (LAK 2024)* (pp. 399–402). SoLAR.

T3 Hagood, D., Spikol, D., Nohr, L., **Echeverria, V.,** Martinez-Maldonado, R., Nieto, G., Srivastava, N., Arslan, A., Curkurova, M., Di Mitri, D., Emerson, A., Giannakos, M., Ochoa, X., & Wang, Y. (2024, March). Design-based research methodology: Going deeper than methods in multimodal learning analytics. In *Companion Proc. of the 14th International Conference on Learning Analytics & Knowledge* (pp. 386–390). SoLAR.

T2 **Echeverria, V.,** Lawrence, L., Tsai, Y., Singh, S., Fernandez-Nieto, G., Martinez-Maldonado, R. (2021). A tutorial on Data Storytelling for Learning Analytics Dashboards – to be held at the 11th International Conference on Learning Analytics and Knowledge, LAK 2021.

T1 Martinez-Maldonado, R., **Echeverria, V.,** Prieto, L. P., Rodriguez-Triana, M. J., Spikol, D., Curukova, M., ... & Worsley, M. (2018). 2nd CrossMMLA: Multimodal learning analytics across physical and digital spaces – held at the 8th International Conference on Learning Analytics and Knowledge, LAK 2018.

GRADUATE SUPERVISION AND MENTORSHIP

SUPERVISION | CO-SUPERVISION

HDR (PhD)

Dipon Talukder	Supervisor	RMIT University	2027
Anh Khoa Huynh	Supervisor	RMIT University	2026
Anh Duy Huynh	Supervisor	RMIT University	2026
Linxuan Zhao	Co-supervisor	Monash University	2023–2025

Riordan Alfredo	Co-supervisor	Monash University	2022–2025

HDR (Masters)

Aryan Shah	Co-supervisor	HCC-FIT, Monash University	2025
Abeer Alshehri	Main supervisor	Monash University	2024-S2–2025-S1
Qiaoyu Deng	Co-supervisor	Monash University	2024-S2–2025-S1
Eleanor Lim	Co-supervisor	Monash University	2024-S2–2025-S1
Vu Dang	Co-supervisor	Monash University	2024-S2–2025-S1

Research Assistants & Developers

Sophie Abel	Research Assistant	FIT, Monash University	2023–present
Jie Fan	Developer	FIT, Monash University	2023–present
Ariel Jin	Research Assistant	FIT, Monash University	2022–present
Eddy Calderón	Developer	CTI, ESPOL	2023–present
Kevin Cevallos	Research Assistant	CTI, ESPOL	2023–present
Benito Auria García	Developer	CTI, ESPOL	2021–2022
Alex Velez	Research Assistant	CTI, ESPOL	2021
Ana Nuñez	Research Assistant	Universidad de Cuenca	2021–2022
Valeria Barzola	Research Assistant	CTI, ESPOL	2021
Judith Taranto	Research Assistant	CTI, ESPOL	2021
Harlyn Pichardo	Developer	CTI, ESPOL	2020–2021

MENTORING

HDR (PhD)

Zainab Zaid	HCC-FIT, Monash University	Jan 2025–present
Paola Mejia	ML4E, EPFL	Feb 2024–Jan 2025
Ann-Christin Falls	Bochum University	2023–present
Ariel Jin	FIT, Monash University	Oct 2023–present
Mikaela Milesi	FIT, Monash University	Oct 2023–present
Conrad Borchers	HCII, Carnegie Mellon University	2023–2024
Kexin (Bella) Yang	HCII, Carnegie Mellon University	2020–2024
Gloria Fernandez-Nieto	CIC, UTS	2019–2021

HDR (Masters)

Hongbo Shao	FIT, Monash University	2022-S2, 2023-S1

Research Assistant Interns

Kate Zhao	HCII, Carnegie Mellon University	May–Aug 2020
Katrina Yu	HCII, Carnegie Mellon University	May–Aug 2020
Jennifer Huang	HCII, Carnegie Mellon University	May–Aug 2020

INVITED TALKS & PANELS

Invited Talk: Multimodal Learning Analytics: Capturing Temporal and Embodied Dimensions of Learning
University of Melbourne, Faculty of Education | June 2026

Invited Talk: A Multimodal Learning Analytics Dashboard to Support Teaching-led Reflection.
HCII – Vincent's Aleven lab | Nov, 2023

Invited Panellist: Human-Centred Artificial Intelligence in Education
Empowering Learners in AI 2022 | Dec, 2022

Invited Lecturer: Interdisciplinary Center for Educational Research
University of Duisburg-Essen, Germany
Human-AI Collaboration Analytics | 2021

Invited Speaker
Brownbag Talk – NYU – USA
Human-AI Collaboration Analytics | 2021

Invited Keynote Speaker
27th Int. Conference on Collaboration Technologies and Social Computing
Collab-Tech 2021 – Germany
Human-AI Collaboration Analytics | 2021

Invited Speaker
WiDS: Women in Data Science
Escuela Superior Politécnica del Litoral, Guayaquil, EC
Analysis and Visualization of Multimodal Data in Learning Environments | 2020

Invited Speaker
Computer Science Department
Escuela Superior Politécnica del Litoral, Guayaquil, EC
The role of AI in Learning Environments | 2020

Invited Speaker: Learning Analytics Unconference
Connected Intelligence Centre, University of Technology Sydney (UTS), Sydney, AUS
Group Analytics: Designing and Validating Automated Feedback for Co-located Groups | 2017

Invited Speaker: Health and Tech Forum
Faculty of Health, University of Technology Sydney (UTS), Sydney, AUS
Physical Learning Analytics in Healthcare Simulation Scenarios | 2017

ACADEMIC SERVICES

Editorial Board Member

International Journal of Computer-Supported Collaborative Learning (iJCSCL) | 2025–present
Journal of Behaviour and Information Technology (BIT) | 2025–present

Guest Editor

British Journal of Educational Technologies (BJET) | 2025-present

Associate Chair

ACM Conference on Human Factors in Computing Systems (CHI) | 2023–2026
ACM Computer Supportive Cooperative Work (CSCW) | 2023–2024

Senior Reviewer

British Journal of Educational Technologies (BJET) | 2020–present
International Journal of Human-Computer Interaction | 2024
Journal of Computer-Assisted Learning (JCAL) | 2024–present
Journal of Learning Analytics | 2020–present
IEEE Transactions on Learning Technologies | 2020–2022
ACM Conference on Human Factors in Computing Systems (CHI) | 2019–present
International Journal of Artificial Intelligence in Education | 2025

Program Committee Member

International Conference of Artificial Intelligence in Education (AIED) | 2026
International Conference of Information and Communication Technologies | 2020
International Learning Analytics and Knowledge (LAK) Conference | 2019–present
International Conference on Technologies Trends | 2018
IEEE Ecuador Technical Chapters Meeting | 2018
International Conference in Multimodal Interaction (ICMI) | 2018–present
Asia-Pacific Conference on Computer Aided System Engineering (APCASE) | 2015

Other Services

Web Chair, Computer Supported Cooperative Work (CSCW) international conference | 2022–2024
Learning Analytics Senior Advisory Group | 2022
Student Volunteer, LAK 18, Sydney, AUS | 2018

PROFESSIONAL MEMBERSHIPS / AFFILIATIONS

IEEE Computational Intelligence Society | 2017–present
IEEE Women in Engineering | 2017–present
Society for Learning Analytics Research (SoLAR) | 2017–present
Institute of Electrical and Electronics Engineers (IEEE) | 2016–present
Association for Computing Machinery (ACM) | 2014–present